Lab 3: DHCP - NAT - IPTABLES

Lecture

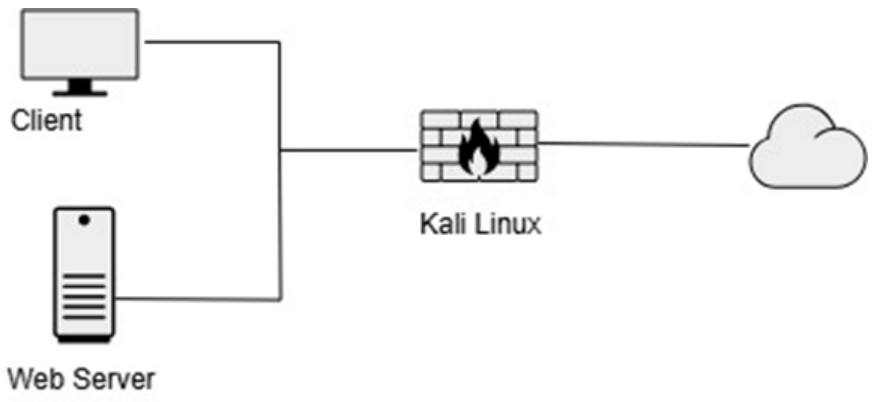
1. Objective

The objective of this lab is to configure a virtual network environment in VMware Workstation using three virtual machines: Kali Linux (gateway), Client, and Web Server. The Kali Linux VM will provide DHCP services on a Host-Only network (VMnet2), enable NAT for internet access, and apply iptables firewall rules for security. Specific tasks include:

- Setting up a DHCP server on Kali to assign IPs in the range 192.168.100.10–50 to Client and Web Server.
- Configuring NAT on Kali to allow Host-Only VMs to access the internet via the NAT interface (192.168.35.129).
- Implementing iptables rules to allow web access from NAT to Host-Only, prevent ICMP floods, and block SSH access from the NAT interface.
- Configuring port forwarding to allow NAT-side access to the Web Server's HTTP service.
- Documenting the process with figures, including a network topology diagram.

2. Environment

- **Kali Linux (Gateway)**:
  - Operating System: Kali Linux.
  - Interfaces: eth0 (Host-Only, static IP 192.168.100.1, VMnet2), eth1 (NAT, DHCP-assigned 192.168.35.129/24, VMnet8).
  - Software: isc-dhcp-server, iptables, iptables-persistent, Apache2 (for testing).
- **Client**:
  - Operating System: Generic Linux (e.g., Ubuntu).
  - Interface: eth0 (Host-Only, VMnet2, DHCP-assigned IP in 192.168.100.10–50).
- **Web Server**:
  - Operating System: Generic Linux (e.g., Ubuntu with Apache2).
  - Interface: eth0 (Host-Only, VMnet2, DHCP-assigned IP in 192.168.100.10–50, e.g., 192.168.100.20).
  - Software: Apache2, iptables, iptables-persistent.
- **Network Environment**: VMware Workstation, Host-Only (VMnet2) for internal network, NAT (VMnet8) for internet access. Connected via DHCP, HTTP (port 80), SSH (port 22), and ICMP.



network_topology.png

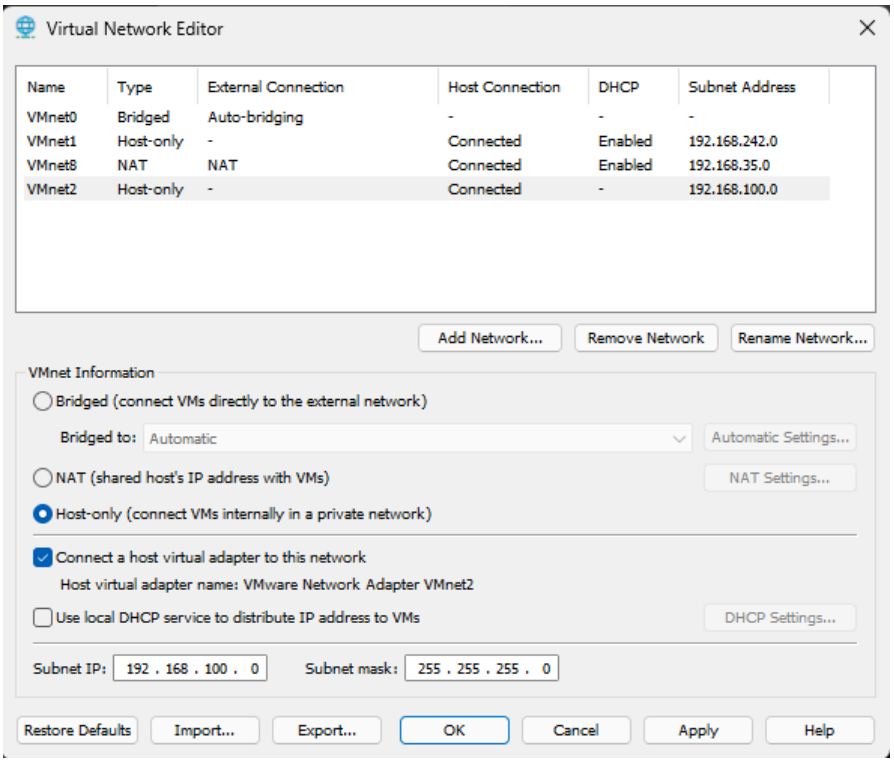
3. Implementation Steps

3.1 Install and Configure VMware Network

Objective: Set up a Host-Only network (VMnet2) and configure VM network adapters.

Step 3.1.1: Create Host-Only Network

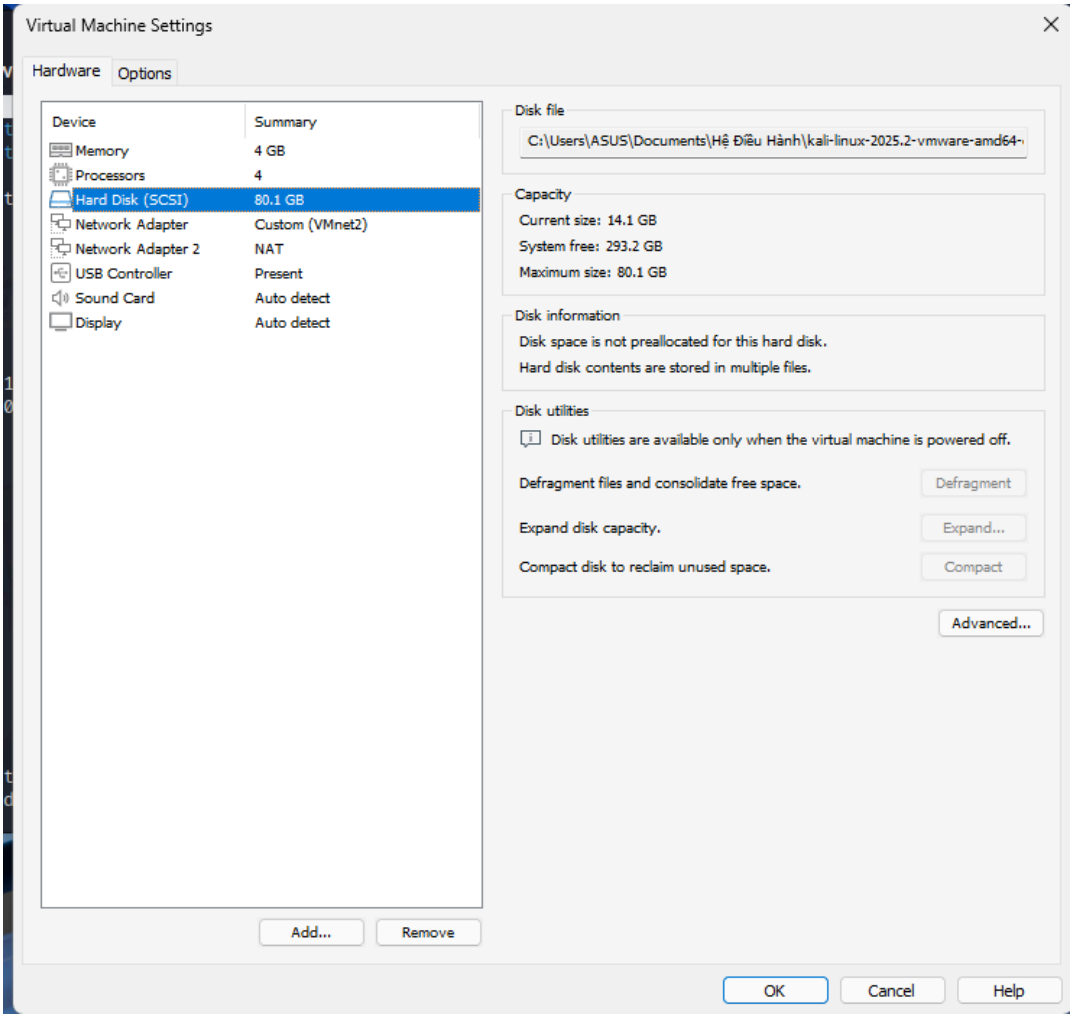
- **Command**:
  - Open VMware Workstation > Edit > Virtual Network Editor.
  - Add Network: VMnet2, set to Host-Only.
  - Uncheck “Use local DHCP service to distribute IP address to VMs”.
- **Purpose**: Creates an isolated network for Client and Web Server, with Kali providing DHCP services.



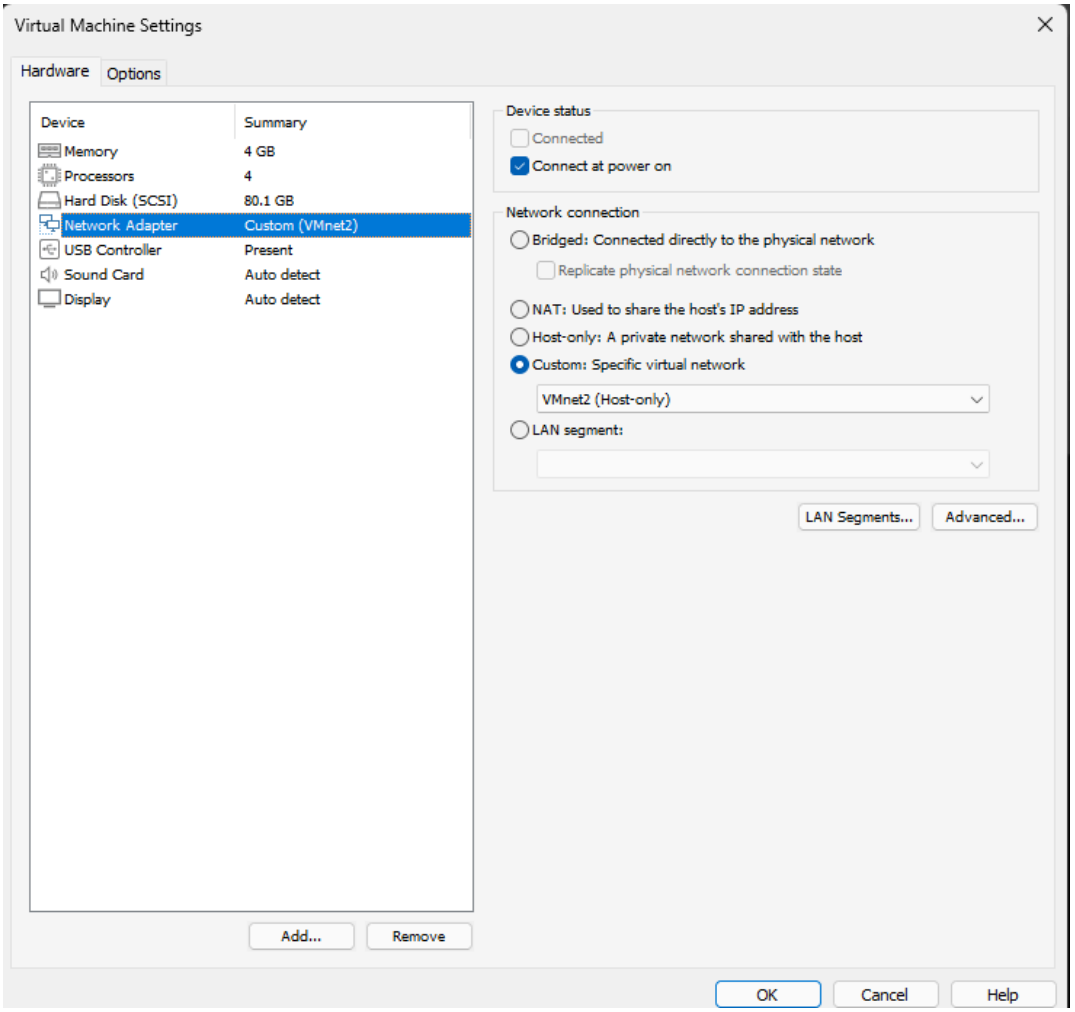
VMware_VMnet2_Config.png

Step 3.1.2: Configure VM Network Interfaces

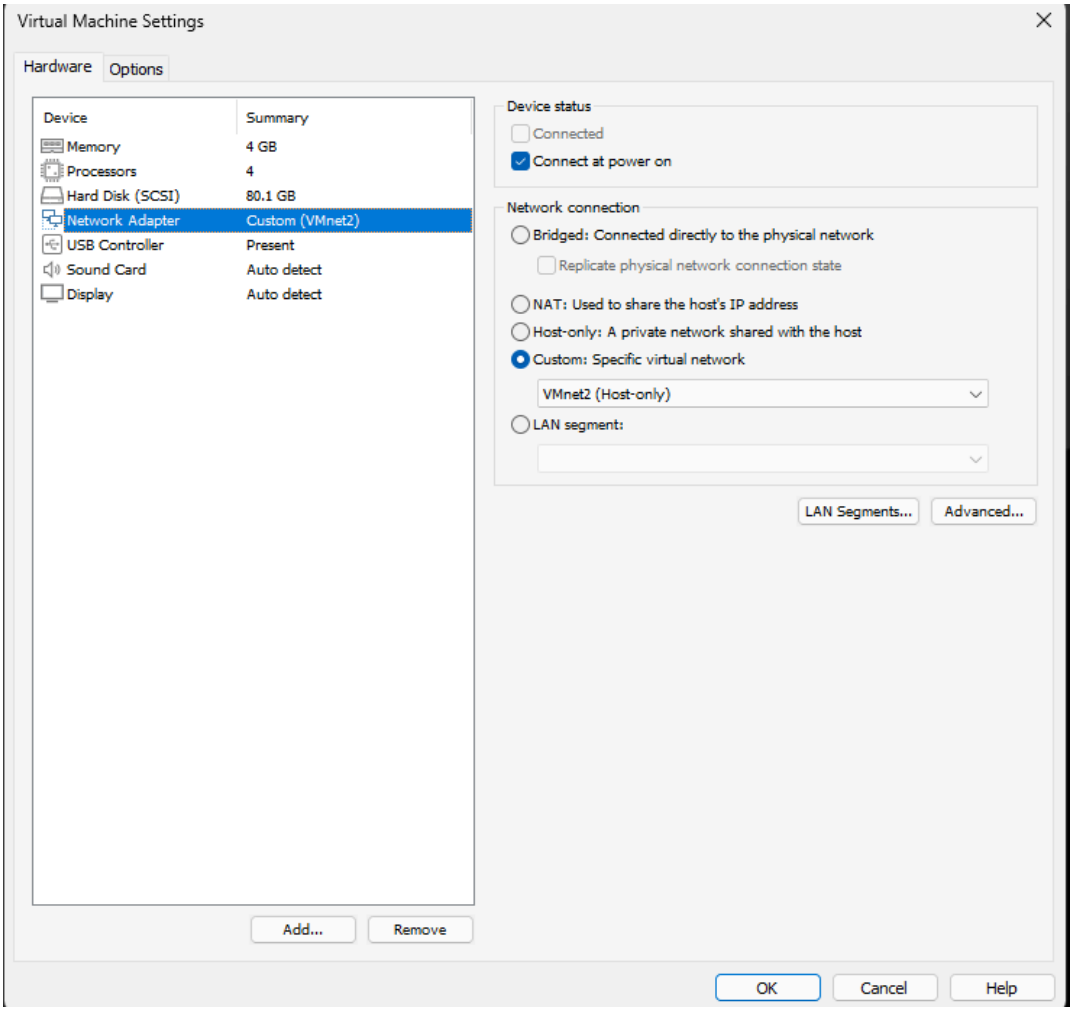
- **Command**:
  - Kali: Adapter 1 (Custom: VMnet2), Adapter 2 (NAT).
  - Client/Web Server: Adapter 1 (Custom: VMnet2).
- **Purpose**: Assigns Host-Only and NAT interfaces to VMs for proper connectivity.



Kali_Network_Adapters.png



Client_Network_Adapter.png



WebServer_Network_Adapter.png

3.2 Configure Kali Linux (Gateway)

Objective: Set up DHCP, NAT, and iptables on Kali Linux to act as a gateway.

Step 3.2.1: Identify Interfaces

- **Command**:

```
ip a
```

- **Purpose**: Confirms interface assignments (eth0: Host-Only, eth1: NAT with 192.168.35.129/24).
- **Output**: Shows eth0 (no IPv4 initially) and eth1 (192.168.35.129/24).

Step 3.2.2: Set Static IP on Host-Only Interface

- **Command**:
  Append at the end:

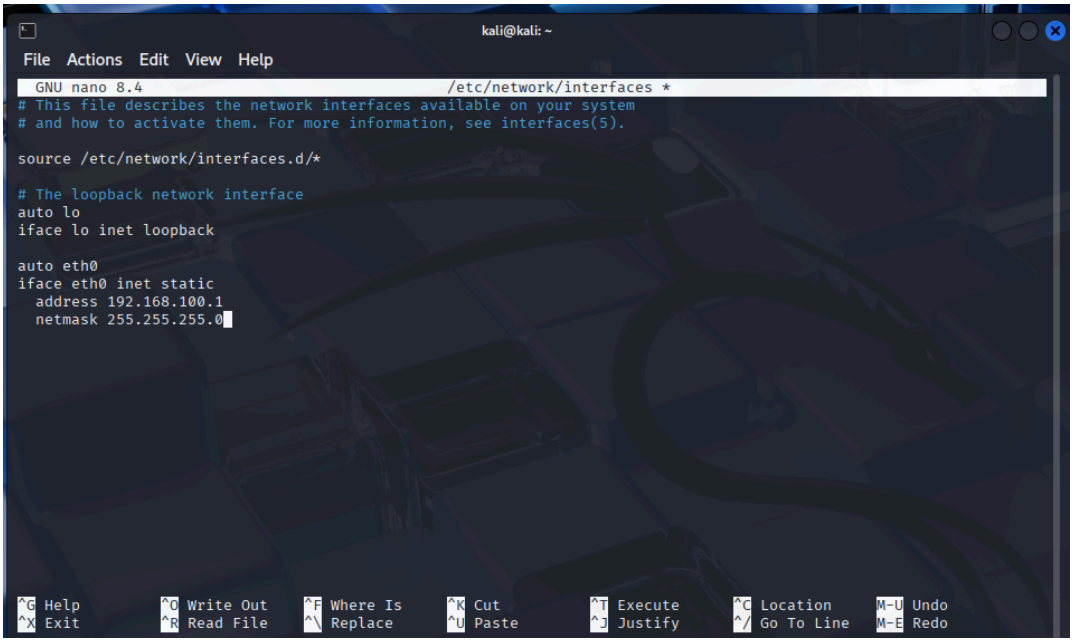
```
sudo nano /etc/network/interfaces
```

```
auto eth0
iface eth0 inet static
```

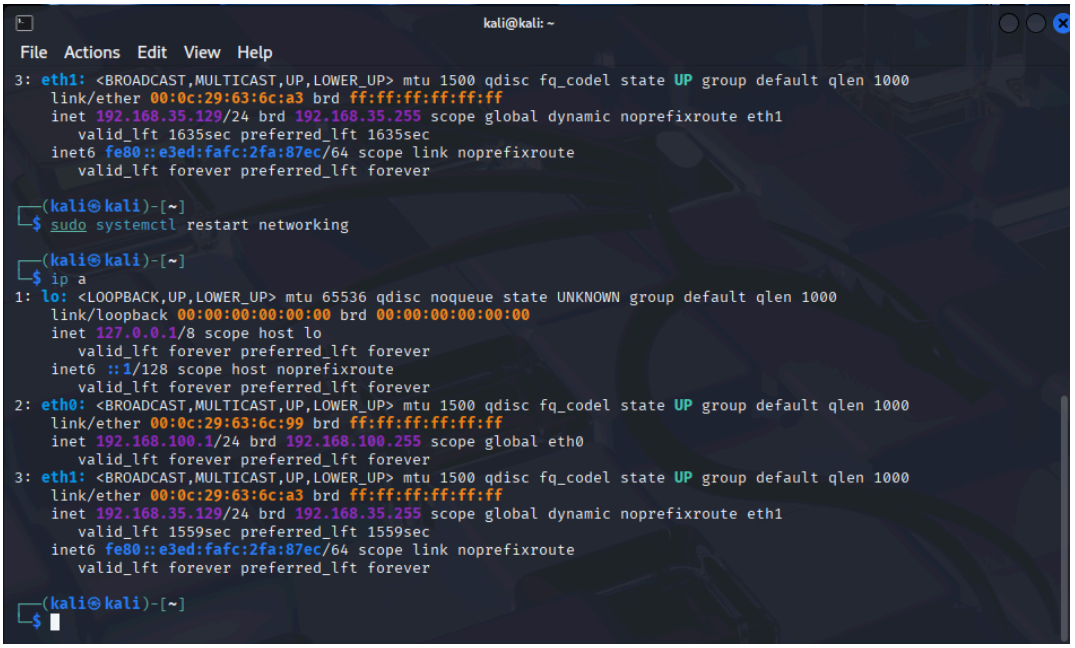
```
  address 192.168.100.1
  netmask 255.255.255.0
```

```
sudo systemctl restart networking
```

- **Purpose**: Assigns static IP 192.168.100.1 to eth0 for DHCP and gateway functionality.



Kali_eth0_IP.png



Kali_eth0_IP.png

Step 3.2.3: Install and Configure DHCP Server

- **Command**:
  Append at the end:

Set (replace or add):

```
sudo apt update && sudo apt install isc-dhcp-server
sudo nano /etc/dhcp/dhcpd.conf
```

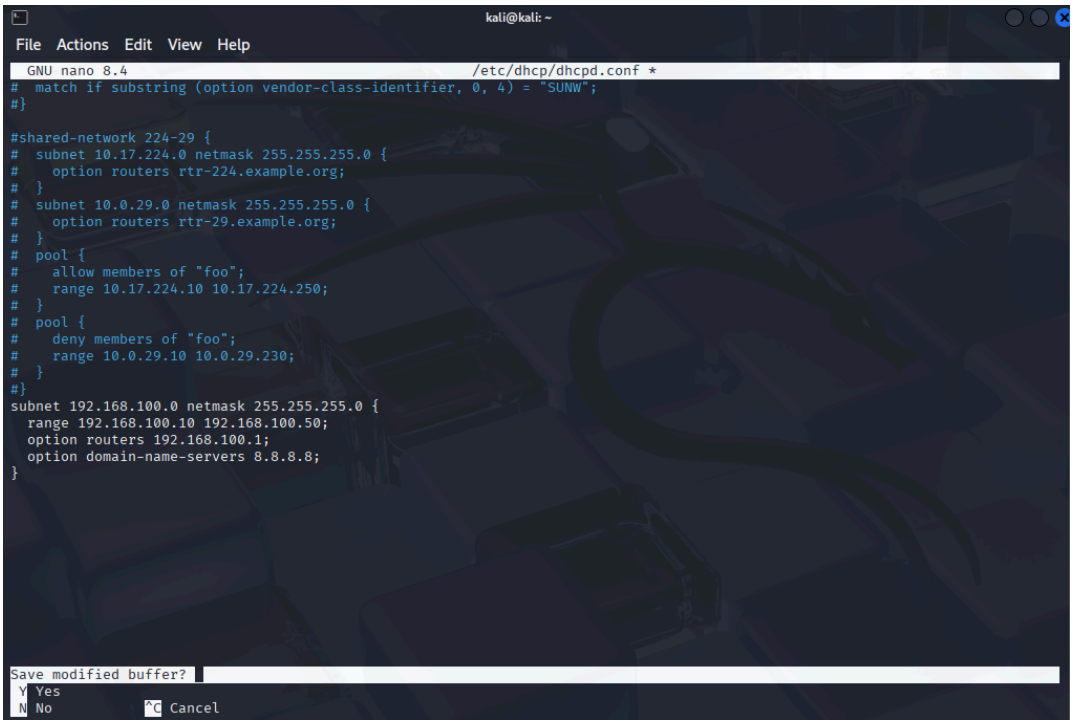
```
subnet 192.168.100.0 netmask 255.255.255.0 {
   range 192.168.100.10 192.168.100.50;
   option routers 192.168.100.1;
   option domain-name-servers 8.8.8.8;
}
```

```
sudo nano /etc/default/isc-dhcp-server
```

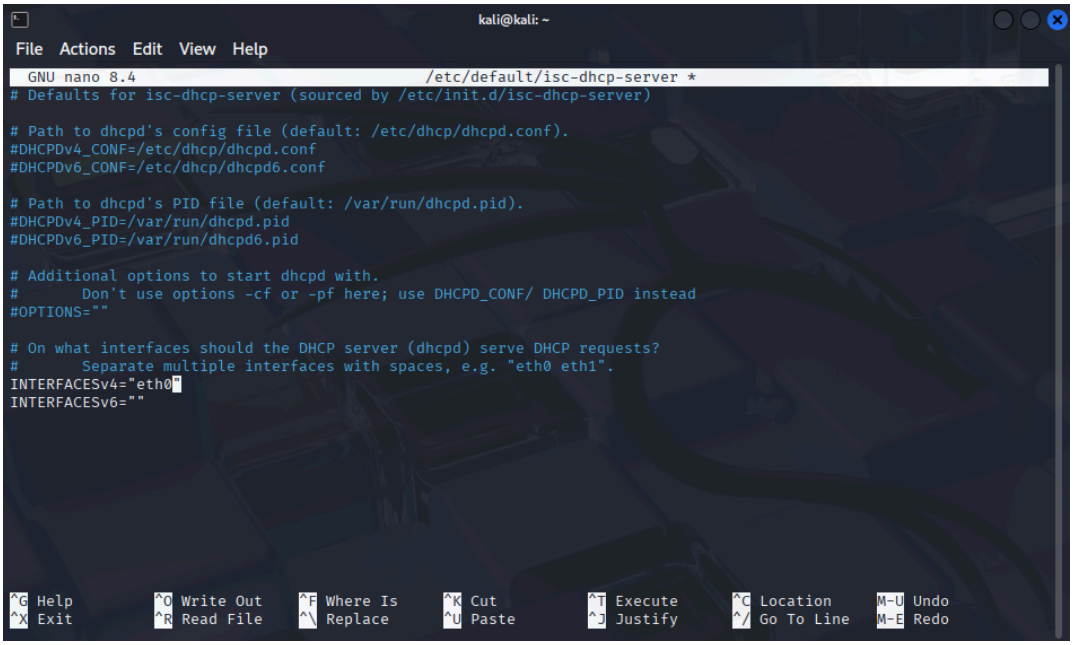
```
INTERFACESv4="eth0"
```

```
sudo systemctl restart isc-dhcp-server
sudo systemctl status isc-dhcp-server
```

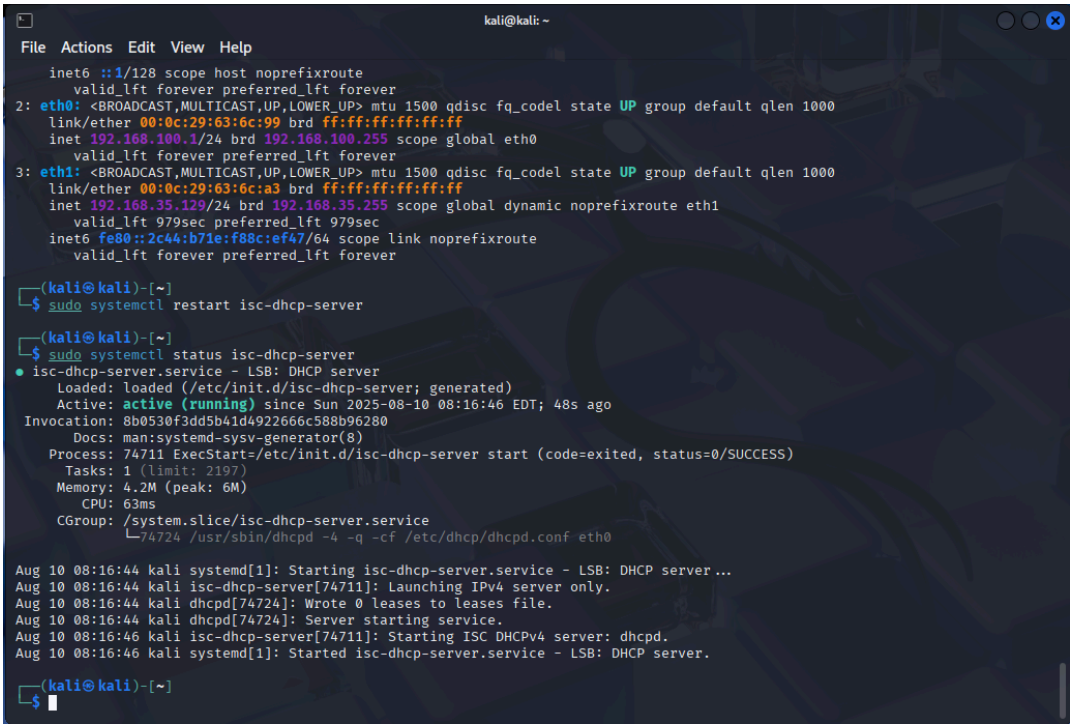
- **Purpose**: Configures DHCP server to assign IPs to Client and Web Server on the Host-Only network.



Kali_DHCP_conf.png



Kali_default_interface.png



Kali_DHCP_Status.png

Step 3.2.4: Enable IP Forwarding

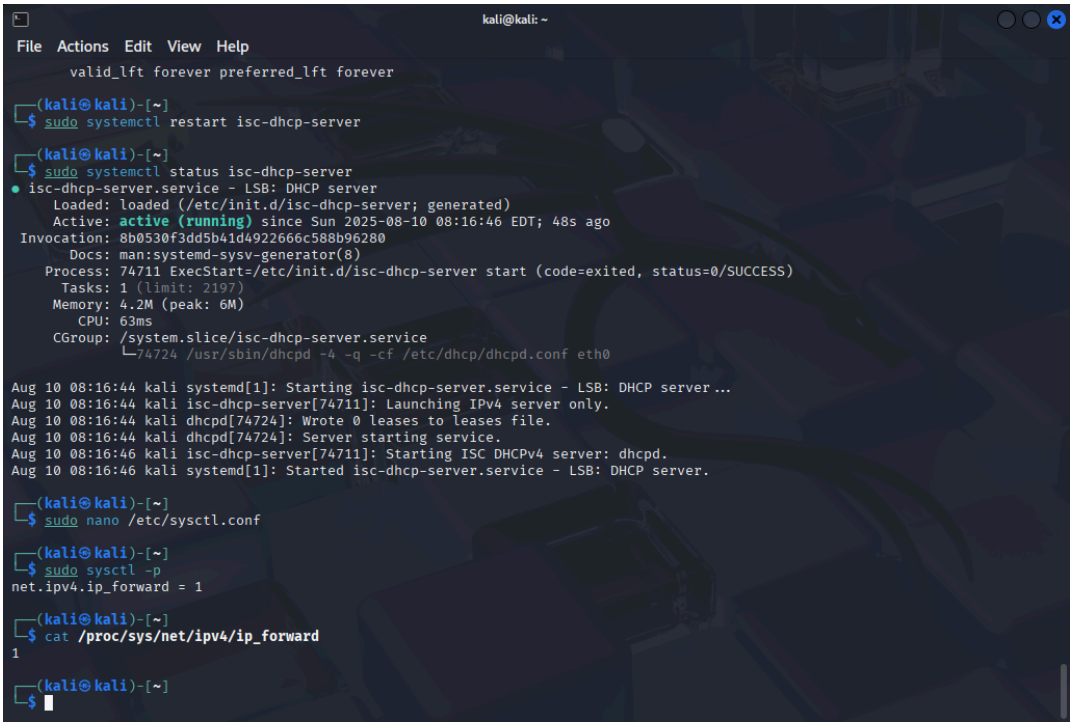
- **Command**:
  Uncomment:

```
sudo nano /etc/sysctl.conf
```

```
net.ipv4.ip_forward=1
```

```
sudo sysctl -p
```

- **Purpose**: Enables packet routing between eth0 and eth1 for NAT functionality.



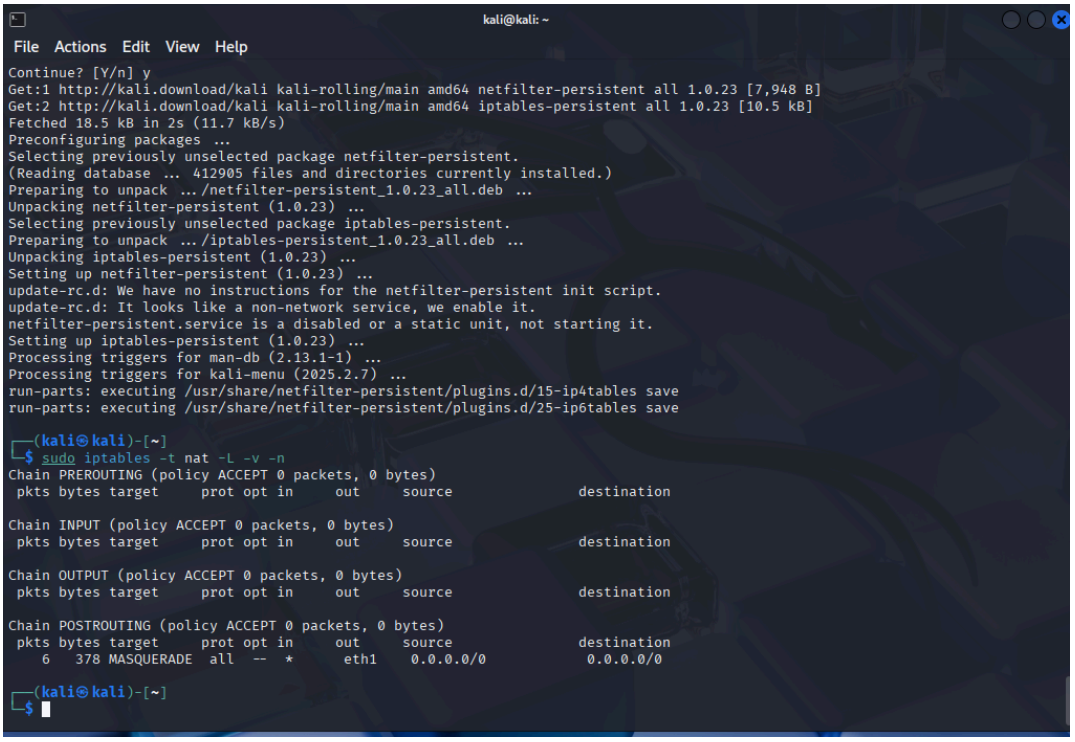
Kali_IP_Forwarding.png

Step 3.2.5: Set Up NAT with iptables

- **Command**:

```
sudo iptables -t nat -A POSTROUTING -o eth1 -j MASQUERADE
sudo apt install iptables-persistent
sudo netfilter-persistent save
```

- **Purpose**: Configures NAT to allow Host-Only VMs to access the internet via eth1 (192.168.35.129).



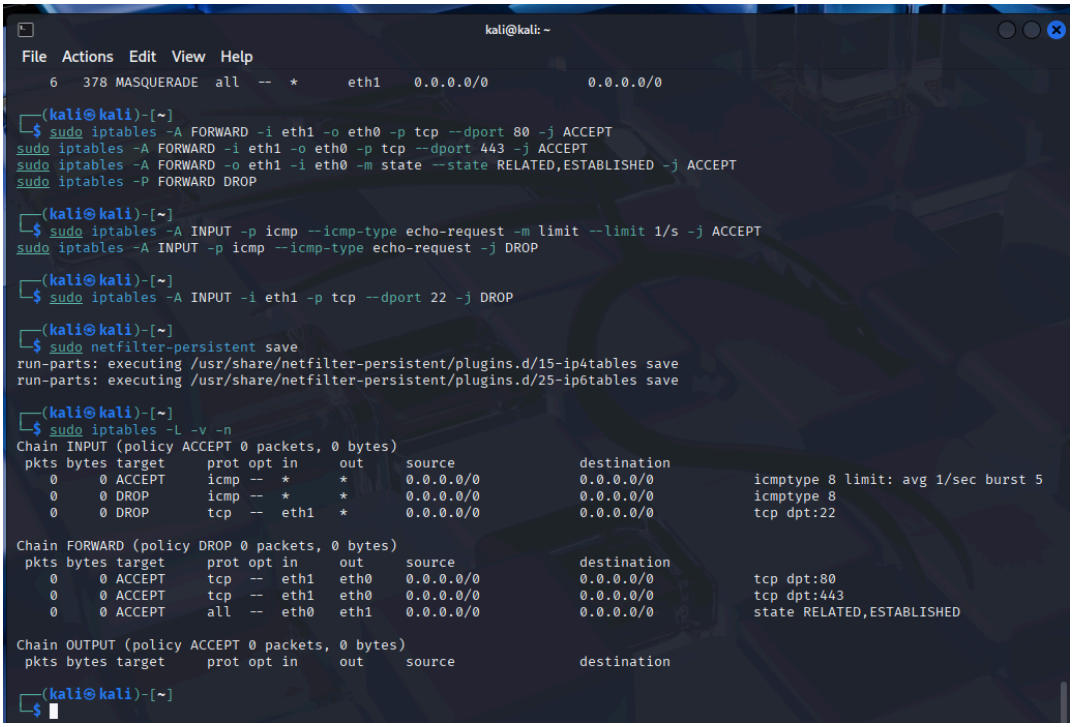
Kali_IPTables_NAT.png

Step 3.2.6: Basic Firewall Rules

- **Command**:

```
sudo iptables -A FORWARD -i eth1 -o eth0 -p tcp --dport 80 -j ACCEPT
sudo iptables -A FORWARD -i eth1 -o eth0 -p tcp --dport 443 -j ACCEPT
sudo iptables -A FORWARD -o eth1 -i eth0 -m state --state RELATED,ESTABLISHED -j ACCEPT
sudo iptables -P FORWARD DROP
sudo iptables -A INPUT -p icmp --icmp-type echo-request -m limit --limit 1/s -j ACCEPT
sudo iptables -A INPUT -p icmp --icmp-type echo-request -j DROP
sudo iptables -A INPUT -i eth1 -p tcp --dport 22 -j DROP
sudo netfilter-persistent save
```

- **Purpose**: Configures firewall to allow web traffic (HTTP/HTTPS) from NAT to Host-Only, limit ICMP to prevent floods, and block SSH from the NAT interface.



Kali_Firewall_Rules.png

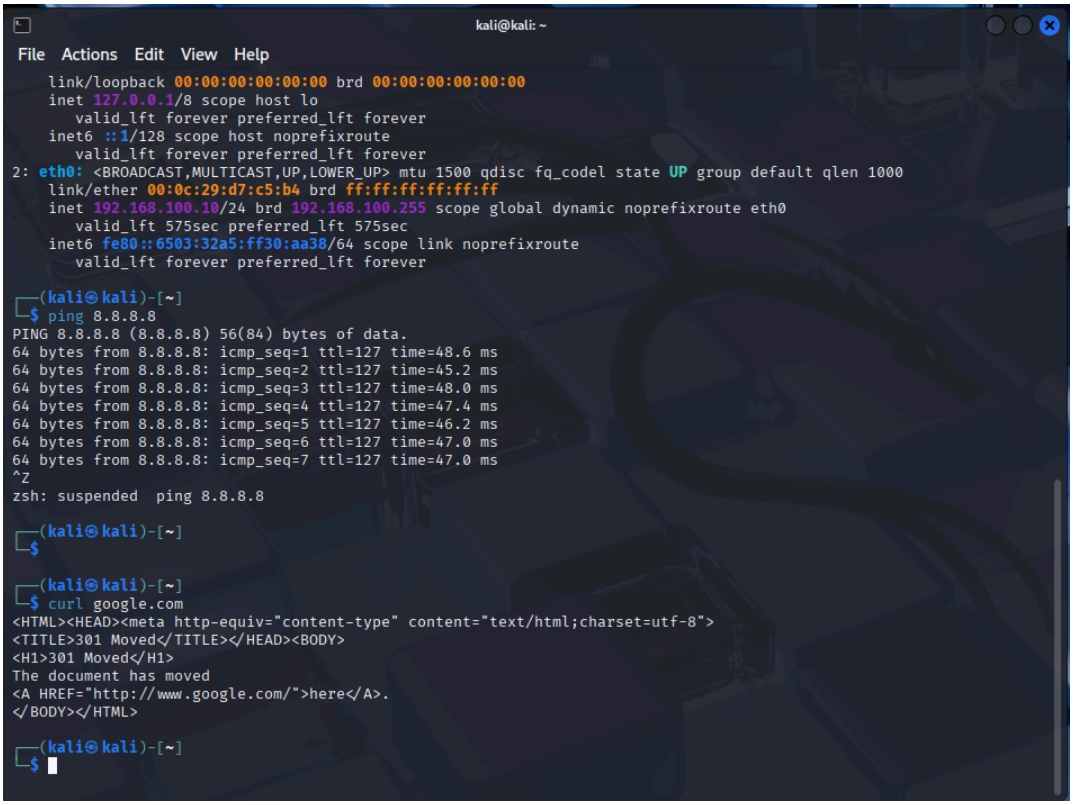
3.3 Configure Client

Objective: Verify DHCP assignment and internet access on the Client VM.

- **Command**:

```
ip a
ping 8.8.8.8
```

- **Purpose**: Confirms the Client receives a DHCP-assigned IP (192.168.100.10–50) and can access the internet via Kali's NAT.



Client_Ping.png

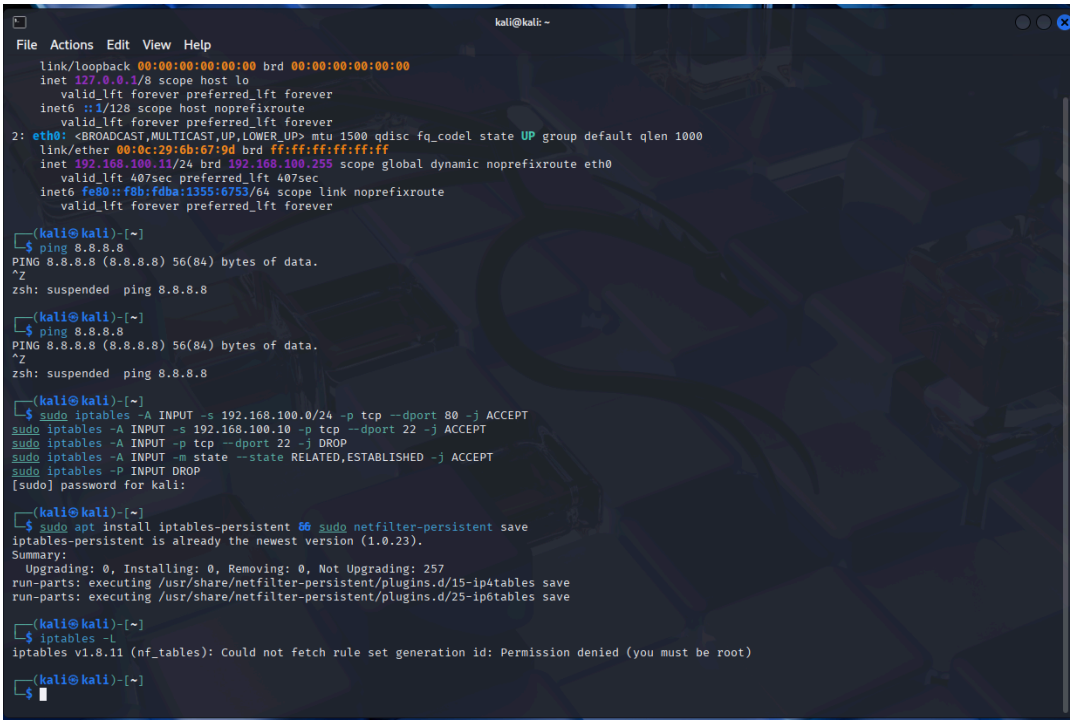
3.4 Configure Web Server

Objective: Set up the Web Server with HTTP service and restricted SSH access.

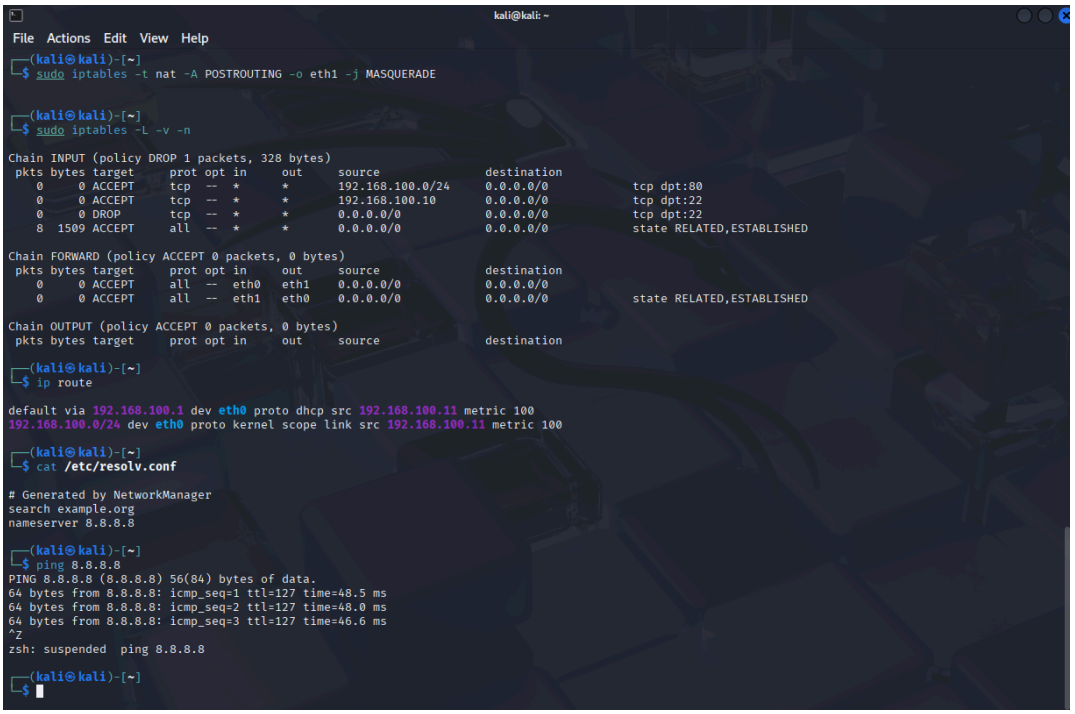
- **Command**:

```
ip a
ping 8.8.8.8
sudo apt install apache2 && sudo systemctl start apache2
sudo iptables -A INPUT -s 192.168.100.0/24 -p tcp --dport 80 -j ACCEPT
sudo iptables -A INPUT -s 192.168.100.10 -p tcp --dport 22 -j ACCEPT
sudo iptables -A INPUT -p tcp --dport 22 -j DROP
sudo iptables -A INPUT -m state --state RELATED,ESTABLISHED -j ACCEPT
sudo iptables -P INPUT DROP
sudo apt install iptables-persistent
sudo netfilter-persistent save
```

- **Purpose**: Configures Apache for HTTP access and restricts SSH to the Client (e.g., 192.168.100.10).



WebServer_IPTables.png



WebServer_Ping.png

3.5 Advanced Part: Port Forwarding

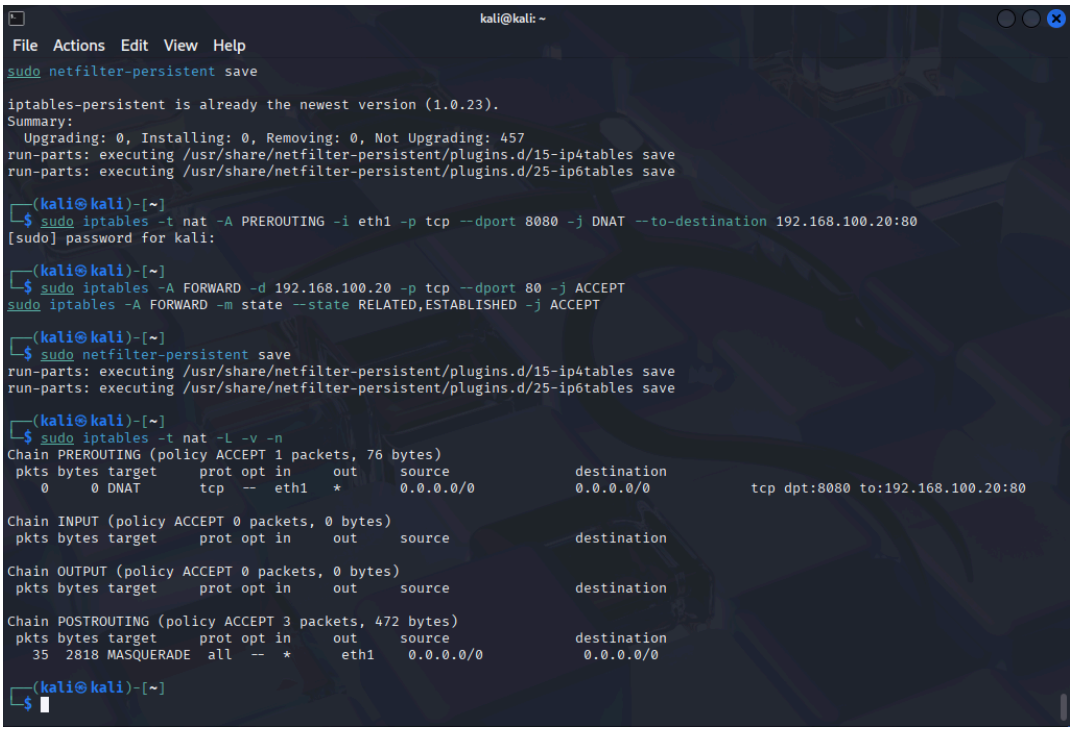
Objective: Enable NAT-side access to the Web Server's HTTP service via port forwarding.

- **Command**:

```
sudo iptables -t nat -A PREROUTING -i eth1 -p tcp --dport 8080 -j DNAT --to-destination 192.168.1
00.20:80
sudo iptables -A FORWARD -d 192.168.100.20 -p tcp --dport 80 -j ACCEPT
```

```
sudo iptables -A FORWARD -m state --state RELATED,ESTABLISHED -j ACCEPT
sudo netfilter-persistent save
```

- **Purpose**: Forwards traffic from Kali's NAT interface (192.168.35.129:8080) to the Web Server (192.168.100.20:80).



PortForward_Rules.png